

# Secure Samba Sharing with Real-Time FIM Auditing and Active ClamAV Response via Wazuh

## Architecture & procedure guide

This document describes implementing a reactive security posture on a home file server (Raspberry Pi 8GB running Debian Trixie). The goal is to couple Wazuh's real-time File Integrity Monitoring (FIM) with the ClamAV antivirus engine, to automatically and immediately scan any file dropped onto a Samba share.

## 1. Solution Architecture

- **Server:** Raspberry Pi (8 GB RAM) — OS: Debian Trixie
- **Key services:** Wazuh Manager, ClamAV Daemon (clamd), Samba
- **Test client:** custom Debian workstation (Kali/Debian) using `smbclient`
- **Reactive logic:** SMB file drop → FIM detection (Wazuh Syscheck) → Active Response trigger (server) → ClamAV scan → custom log & alert

## 2. Secure Samba Share Configuration

The shared directory was configured to strictly restrict access to a single dedicated group ( `partageur` ), with restrictive creation masks:

```
[Partage-Commun]
comment = Partage Commun Securise
path = /home/partage_commun
browseable = yes
read only = no
guest ok = no
valid users = @partageur
force group = partageur
create mask = 0770
directory mask = 0770
```

## 3. Firewall Configuration (UFW)

To allow access to the Samba share while keeping network filtering in place, the NetBIOS and SMB Direct listening ports were explicitly allowed:

```
sudo ufw allow 139/tcp
sudo ufw allow 445/tcp
sudo ufw reload
```

## 4. Installing and Configuring ClamAV

To avoid overloading the Raspberry Pi's memory with repeated scans, ClamAV was deployed in daemonized mode ( `clamd` ). The `clamav` user was added to the local `partageur` group to guarantee read access to the shared data without privilege escalation:

```
sudo apt install clamav clamav-daemon -y
sudo usermod -aG partageur clamav
sudo freshclam
sudo systemctl enable clamav-daemon --now
```

## 5. Linking ClamAV with Wazuh (FIM Configuration)

Wazuh's integrity-audit module (Syscheck) was configured to synchronously monitor, in real time, the root directory of the Samba shares:

```
<directory realtime="yes">/home/partage_commun</directory>
```

## 6. Active Response Script ( `clamav-scan.sh` )

The response script was custom-built within Wazuh's directory tree. It intercepts the JSON payload sent by the Manager, extracts the absolute file path via `jq`, and triggers a quick local scan via the ClamAV daemon:

```
#!/bin/bash
read -r INPUT
FILE_PATH=$(echo "$INPUT" | jq -r '.parameters.alert.syscheck.path')
LOG_FILE="/var/ossec/logs/active-responses.log"

if [ "$FILE_PATH" != "null" ] && [ -f "$FILE_PATH" ]; then
    SCAN_RESULT=$(clamscan --no-summary "$FILE_PATH")
    if echo "$SCAN_RESULT" | grep -q "FOUND"; then
        echo "$(date) - [MALWARE ALERT] Infected file detected: $FILE_PATH - Result: $SCAN_RESULT" >> "$LOG_FILE"
    fi
fi
```

For hardening purposes, permissions on the script are restricted to the Wazuh security daemon:

```
sudo chown root:wazuh /var/ossec/active-response/bin/clamav-scan.sh
sudo chmod 750 /var/ossec/active-response/bin/clamav-scan.sh
```

## 7. Automated Response Configuration ( `ossec.conf` )

The final integration links Syscheck rule 554 (file-added detection) to the scan script. To guarantee execution on the local node hosting the storage, the location target is set to `server` :

```
<command>
  <name>clamav-scan</name>
  <executable>clamav-scan.sh</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <command>clamav-scan</command>
  <location>server</location>
  <rules_id>554</rules_id>
</active-response>
```

## 8. Validation & Testing (Purple Team Approach)

The pipeline was validated by injecting the standardized EICAR test string from the Samba client environment:

```
smb: \> ! echo 'X50!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > virus_test.txt
smb: \> put virus_test.txt
```

The system's immediate reactivity was confirmed by the synchronous write of the antivirus verdict to the Active Response log:

```
Sat Jul 18 16:13:49 UTC 2026 - [MALWARE ALERT] Infected file detected: /home/partage_commun/virus_test.txt - Result:
/home/partage_commun/virus_test.txt: Eicar-Signature FOUND
```